

Experiment 1: LINUX NETWORKING COMMANDS

1: ip addr show

```
anexam@fedora:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP>; mtu 65536 qdisc noqueue state UNKNOWN group
default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP>; mtu 1500 qdisc fq_codel state
UP
group default qlen 1000
link/ether 7c:57:58:38:e6:24 brd ff:ff:ff:ff:ff:ff
inet 172.16.8.186/22 brd 172.16.11.255 scope global noprefixroute enp1s0
valid_lft forever preferred_lft forever
inet6 fe80::cea4:ebab:7815:e003/64 scope link noprefixroute
valid_lft forever preferred_lft forever
```

2: if config enp1s0

```
anexam@fedora:~$ ifconfig enp1s0
enp1s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>; mtu 1500
inet 172.16.8.186 netmask 255.255.252.0 broadcast 172.16.11.255
inet6 fe80::cea4:ebab:7815:e003 prefixlen 64 scopeid 0x20<link>;
ether 7c:57:58:38:e6:24 txqueuelen 1000 (Ethernet)
RX packets 32048 bytes 16306284 (15.5 MiB)
RX errors 0 dropped 40 overruns 0 frame 0
TX packets 10040 bytes 4676069 (4.4 MiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

3: ping google.com

```
anexam@fedora:~$ ping google.com
PING google.com (142.251.222.206) 56(84) bytes of data.
64 bytes from pnmaaa-as-in-f14.1e100.net (142.251.222.206): icmp_seq=1 ttl=119 time=3.82
ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=2 ttl=119 time=4.63 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=3 ttl=119 time=4.10 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=4 ttl=119 time=4.94 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=5 ttl=119 time=4.10 ms
64 bytes from hkg07s55-in-f14.1e100.net (142.251.222.206): icmp_seq=6 ttl=119 time=4.15 ms
^C
--- google.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
```

rtt min/avg/max/mdev = 3.817/4.289/4.942/0.379 ms

4: traceroute google.com

anexam@fedora:~\$ traceroute google.com

traceroute to google.com (142.250.205.110), 30 hops max, 60 byte packets

```
1 _gateway (172.16.8.1) 0.319 ms 0.289 ms 0.274 ms
2 137.97.163.129 (137.97.163.129) 1.772 ms 4.010 ms 3.277 ms
3 * * *
4 172.16.12.122 (172.16.12.122) 5.491 ms 172.31.5.65 (172.31.5.65) 5.108 ms
  172.16.12.122 (172.16.12.122) 4.001 ms
5 172.16.12.64 (172.16.12.64) 5.770 ms 172.16.12.122 (172.16.12.122) 4.412 ms
  209.85.175.48 (209.85.175.48) 5.065 ms
6 209.85.175.48 (209.85.175.48) 4.763 ms * 5.107 ms
7 * 142.251.55.28 (142.251.55.28) 3.844 ms *
8 142.250.208.152 (142.250.208.152) 19.784 ms 142.251.55.120 (142.251.55.120) 5.329 ms
  142.250.224.6 (142.250.224.6) 4.823 ms
9 142.251.230.53 (142.251.230.53) 5.139 ms 74.125.252.91 (74.125.252.91) 5.433 ms
  142.250.208.152 (142.250.208.152) 4.522 ms
10 142.251.230.71 (142.251.230.71) 4.649 ms 209.85.248.181 (209.85.248.181) 4.721 ms
   74.125.252.91 (74.125.252.91) 5.665 ms
11 pnmaaa-ao-in-f14.1e100.net (142.250.205.110) 4.995 ms 4.514 ms 4.852 ms
```

5: tracepath

anexam@fedora:~\$ tracepath

Usage

tracepath [options] <destination>;

Options:

-4 use IPv4

-6 use IPv6

-b print both name and ip

-l <length>; use packet <length>;

-m <hops>; use maximum <hops>;

-n no dns name resolution

-p <port>; use destination <port>;

-V print version and exit

<destination>; dns name or ip address

For more details see tracepath(8).

6: netstat -an

unix 3 [] STREAM CONNECTED 8510

@2873e70ecb9b9273/bus/dbus-broker-lau/system

unix 2 [] STREAM 12638 @printer-applet-lock-user-anexam

unix 3 [] STREAM CONNECTED 10683

@658298809eae8286/bus/systemd/bus-api-system

```

unix 3 [ ] STREAM CONNECTED 3767
@77b8aae41221e750/bus/systemd-resolve/bus-api-resolve
unix 3 [ ] STREAM CONNECTED 761
@a40aefa74a5a0e65/bus/systemd/bus-system
unix 2 [ ACC ] STREAM LISTENING 10669
@ISCSIADM_ABSTRACT_NAMESPACE
unix 3 [ ] STREAM CONNECTED 6525
@6f7d820d38212fe/bus/systemd-oomd/bus-api-oom
unix 3 [ ] STREAM CONNECTED 10977
@ba45ac851da7673a/bus/dbus-broker-lau/user
Active Bluetooth connections (servers and established)
Proto Destination Source State PSM DCID SCID IMTU OMTU
Security
Proto Destination Source State Channel

```

7: ss -tuln

```

anexam@fedora:~$ ss -tuln
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
udp UNCONN 0 0 127.0.0.54:53 0.0.0.0:*
udp UNCONN 0 0 127.0.0.53%lo:53 0.0.0.0:*
udp UNCONN 0 0 127.0.0.1:323 0.0.0.0:*
udp UNCONN 0 0 0.0.0.0:5353 0.0.0.0:*
udp UNCONN 0 0 0.0.0.0:5355 0.0.0.0:*
udp UNCONN 0 0 0.0.0.0:50662 0.0.0.0:*
udp UNCONN 0 0 [::1]:323 [::]:*
udp UNCONN 0 0 [::]:5353 [::]:*
udp UNCONN 0 0 [::]:5355 [::]:*
udp UNCONN 0 0 [::]:56547 [::]:*
tcp LISTEN 0 4096 127.0.0.54:53 0.0.0.0:*
tcp LISTEN 0 4096 127.0.0.1:631 0.0.0.0:*
tcp LISTEN 0 4096 127.0.0.53%lo:53 0.0.0.0:*
tcp LISTEN 0 4096 0.0.0.0:5355 0.0.0.0:*
tcp LISTEN 0 4096 [::1]:631 [::]:*
tcp LISTEN 0 4096 [::]:5355 [::]:*

```

8: hostname

```

anexam@fedora:~$ hostname
Fedora

```

9: hostnamectl

```

anexam@fedora:~$ hostnamectl
Static hostname: (unset)
Transient hostname: fedora
Icon name: computer-desktop

```

Chassis: desktop 
Machine ID: 5e282386f50d4c36b2128d5c60d3c03f
Boot ID: b4db003a4c2d4ac58c11cf36498661e2
Operating System: Fedora Linux 39 (MATE-Compiz)
CPE OS Name: cpe:/o:fedoraproject:fedora:39
OS Support End: Tue 2024-11-12
OS Support Expired: 1y 8month 1w 1d
Kernel: Linux 6.11.9-100.fc39.x86_64
Architecture: x86-64
Hardware Vendor: HP
Hardware Model: HP Pro Tower 280 G9 PCI Desktop PC
Firmware Version: F.24
Firmware Date: Fri 2023-11-17
Firmware Age: 2y 8month 4d

10: nslookup google.com

```
anexam@fedora:~$ nslookup google.com
Server: 127.0.0.53
Address: 127.0.0.53#53
Non-authoritative answer:
Name: google.com
Address: 142.250.205.110
Name: google.com
Address: 2404:6800:4007:807::200e
```

11: dig google.com

```
anexam@fedora:~$ dig google.com
; <<<>>> DiG 9.18.28 <<<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 29539
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:;, udp: 65494
;; QUESTION SECTION:
;google.com. IN A
;; ANSWER SECTION:
google.com. 273 IN A 142.250.205.110
;; Query time: 1 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Wed Jul 22 16:05:41 IST 2026
;; MSG SIZE rcvd: 55
```

12: host google.com

```
anexam@fedora:~$ host google.com
google.com has address 142.250.205.110
google.com has IPv6 address 2404:6800:4007:807::200e
google.com mail is handled by 10 smtp.google.com.
```

13: arp

```
anexam@fedora:~$ arp
Address HWtype HWaddress Flags Mask Iface
_gateway ether 7c:5a:1c:cf:be:45 C enp1s0
172.16.10.177 ether 88:ae:dd:14:8a:26 C enp1s0
172.16.10.180 ether 88:ae:dd:14:75:ec C enp1s0
172.16.10.183 ether 88:ae:dd:15:ed:5c C enp1s0
172.16.10.132 ether f8:ed:fc:5c:0b:9d C enp1s0
```

14: ip neigh

```
anexam@fedora:~$ ip neigh
172.16.8.1 dev enp1s0 lladdr 7c:5a:1c:cf:be:45 REACHABLE
172.16.10.177 dev enp1s0 lladdr 88:ae:dd:14:8a:26 REACHABLE
172.16.10.180 dev enp1s0 lladdr 88:ae:dd:14:75:ec REACHABLE
172.16.10.183 dev enp1s0 lladdr 88:ae:dd:15:ed:5c REACHABLE
172.16.10.132 dev enp1s0 lladdr f8:ed:fc:5c:0b:9d REACHABLE
172.16.10.191 dev enp1s0 lladdr 88:ae:dd:15:eb:f3 REACHABLE
172.16.8.171 dev enp1s0 lladdr 88:ae:dd:15:eb:ed STALE
172.16.10.194 dev enp1s0 lladdr 88:ae:dd:15:e7:f4 STALE
172.16.10.176 dev enp1s0 lladdr 88:ae:dd:15:ee:a3 STALE
172.16.10.147 dev enp1s0 lladdr f8:ed:fc:5c:09:9b STALE
```

15: ip route

```
anexam@fedora:~$ ip route

default via 172.16.8.1 dev enp1s0 proto static metric 100
172.16.8.0/22 dev enp1s0 proto kernel scope link src 172.16.8.186 metric 100
```

16: curl

```
anexam@fedora:~$ curl
curl: try 'curl --help' or 'curl --manual' for more information
```

17: wget

```
anexam@fedora:~$ wget
wget: missing URL
Usage: wget [OPTION]... [URL]...
Try 'wget --help' for more options.
```

Or

```
wget https://www.google.com
anexam@fedora:~$ wget https://www.google.com
--2026-07-22 16:29:21-- https://www.google.com/
Resolving www.google.com (www.google.com)... 142.251.152.119, 142.251.153.119,
142.251.154.119, ...
Connecting to www.google.com (www.google.com)|142.251.152.119|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'index.html'
index.html [ &lt;=&gt; ] 81.15K --.-KB/s in 0.01s
2026-07-22 16:29:21 (6.25 MB/s) - 'index.html' saved [83095]
```

18:scp

```
anexam@fedora:~$ scp
usage: scp [-346ABCOpqRrsTv] [-c cipher] [-D sftp_server_path] [-F ssh_config]
[-i identity_file] [-J destination] [-l limit] [-o ssh_option]
[-P port] [-S program] [-X sftp_option] source ... target
```

19:ssh

```
anexam@fedora:~$ ssh
usage: ssh [-46AaCfGgKkMNnqsTtVvXxYy] [-B bind_interface]
[-b bind_address] [-c cipher_spec] [-D [bind_address:]port]
[-E log_file] [-e escape_char] [-F configfile] [-I pkcs11]
[-i identity_file] [-J [user@]host[:port]] [-L address]
[-l login_name] [-m mac_spec] [-O ctl_cmd] [-o option] [-p port]

[-Q query_option] [-R address] [-S ctl_path] [-W host:port]
[-w local_tun[:remote_tun]] destination [command [argument ...]]
```

20:ssh

```
anexam@fedora:~$ ssh
usage: ssh [-46AaCfGgKkMNnqsTtVvXxYy] [-B bind_interface]
[-b bind_address] [-c cipher_spec] [-D [bind_address:]port]
[-E log_file] [-e escape_char] [-F configfile] [-I pkcs11]
[-i identity_file] [-J [user@]host[:port]] [-L address]
[-l login_name] [-m mac_spec] [-O ctl_cmd] [-o option] [-p port]
[-Q query_option] [-R address] [-S ctl_path] [-W host:port]
[-w local_tun[:remote_tun]] destination [command [argument ...]]
```